POWER ENTRY (7-pin XLR NC7MD-LX on the rear panel. two 5 A contacts per rail. 18 AWG to this terminal: +-17 V PSU)



DISTRIBUTION (PTC per wing per rail. 1.6 A hold. 72 V. local branches VA audio. VP phones. VM meters. VL logic)

MASTER FADER (dual 100 mm ALPS 10 kΩ log 100mm dual)

MONO: inverting sum of POST L+R. re-inverter. SOLO LEVEL. follower.

PFL SWITCHER: DG413 (PFL pressed = SW1/SW4. else SW2/SW3. break-before-make) -> CR5 L/R OPA1656 followers

MAIN OUTPUTS: DRV134 -> 1Ω NF in the Sense loop -> mute relay (NO: NC grounds the line) -> XLR mate. pin 2 hot

POWER-ON MUTE: rails > -14 V for ~2.7 s closes all six relays: either rail low opens them at once

PFL DETECT (0.92 V per press vs 0.50 V. 2 x 1N4148 clamp. 1µ debounce) and DG413 logic. bestde the switcher

RUDE SOLO: ~2 Hz flasher. PFL ACTIVE LED. held off through FLOW while no key is pressed

METER R: full-wave rectifier. precision peak / average. follower (TL072). 2 dB ladder. 2 x 10-LED strings

CONTROL ROOM: C/ROOM dual AIOK. followers. DRV134. mute relays. TRS outs

AUX MASTER 5: OPA4612 sum (2ZK || 100p) + re-invert. level. buffer. DRV134 out. AFL

AUX MASTER 6: OPA4612 sum (2ZK || 100p) + re-invert. level. buffer. DRV134 out. AFL

MIX L: summing amp (GCANCEL-referenced). re-inverter. master insert

MIX R: summing amp (GCANCEL-referenced). re-inverter. master insert

GROUND CANCEL BUFFER (feeds every re-inverter: summing amps use the raw GCANCEL line: 100K keeper)

MOTHERLOOM (master in the center: CN1 = left wing. CN2 = right wing: Delta pinout)



REMOTE METER HEADER (for a meter bridge)



HEADPHONES: PHONES AIOK. OPA1612 + LME49600 composite (+9.9 dB). 2R2 out. mute relay. PHONESGND

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EXT INPUT (2-track / DAW) and MIX / EXT monitor select (SW latched = EXT)

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Wires joined by labels: refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.7

Sheet: /	
File: master.kicad_sch	
Title: **Master section — goa Console**	Rev: 0.7
Size: User	Date: 2026-10-19
KiCad 8.0.x	Id: 1/1